Mid Exam Important Questions

Unit 1:

1. Explain the semantic tags of HTML5 and their importance.
2. What is CSS Flexbox? Explain its properties.
3. What is CSS? Explain different types of CSS with example.
4. Differentiate between GET and POST methods in AJAX.
5. Explain the HTTP request-response cycle. How does client-server communication work on the web?
6. Explain the concept of AJAX. How does it work? What are its advantages?

Unit 2:

1. What is NPM? Explain its purpose and significance in NodeJS.
2. What is NodeJS? Discuss its role and significance in server-side JavaScript development.
3. What is a Web Server? Explain key features.
4. Explain Express JS in brief.
5. What is the event loop in NodeJS? Explain how asynchronous, non-blocking I/O works.
6. What are templating engines in NodeJS? Explain server-side rendering with an example (EJS/Pug).

Unit 3:

1. Explain the concept of NoSQL databases. How do they differ from traditional relational databases?
2. What is MongoDB? Discuss its architecture and primary features. Explain CRUD operations in MongoDB with shell commands (insertOne, find/findOne, updateOne, deleteOne).
3. Mention differences between SQL and NoSQL databases.
4. What is a Cookie? How is it handled in Node.js?
5. What is Mongoose? How is it used to define schemas and models in NodeJS?
6. What is body-parser middleware in ExpressJS? How is request body parsed?

Mid Exam Important Questions

Unit 4:

1. What is JSX, and why is it used in React applications?
2. What is ReactDOM? Explain in detail.
3. What is LocalStorage in JavaScript? How does it differ from SessionStorage and cookies?
4. Explain the purpose of the Fetch API in React applications.
5. How does ReactDOM render elements into the DOM?
6. What is a component? Explain types with example. Explain the component lifecycle methods in React (mounting, updating, unmounting).
7. What are React Hooks? Explain any two with examples.
8. What are props in React? How are they different from state? Explain with an example.
9. Explain the React component lifecycle methods (mounting, updating, unmounting) in class components.
10. Explain state management in React. What is Lifting State Up in React? Explain with an example of passing state between components.

Mid Exam Important Questions

Unit 5:

1. What is scaling in cloud? Explain the concepts of horizontal scaling and vertical scaling with examples.
2. Explain Docker in detail.
3. Explain Kubernetes in detail.
4. Explain Virtual Private Cloud (VPC) in detail.
5. What is Kubernetes? Explain its architecture.
6. What is EC2 in Cloud?
7. What is a Docker container? Why use Docker?
8. Explain the architecture of Kubernetes in detail. Describe master node, worker node, pods, and services.
9. What is a Docker image? How is it different from a Docker container? Explain Dockerfile.
10. What is a Virtual Machine (VM)? How does it differ from a Docker container?
11. Explain the concept of Virtual Private Cloud (VPC). What are its components and benefits?
12. Explain the Dockerfile and Docker image concepts. How do you build and run a Docker image?
13. What is Load Balancing in cloud? How does Kubernetes handle it?
14. What is Ethernet and switching in cloud networking? Explain briefly.
15. Explain the concept of containerization. How is it different from virtualization?